AWS API Gateway — Important Interview Questions & Answers

1.What is Amazon API Gateway?
Amazon API Gateway is a fully managed AWS service used to create, publish, secure, monitor, and manage APIs.
It acts as a front door for applications to access backend services such as Lambda, EC2, and other AWS services.
Simple ga gurthupettuko:
API Gateway ante applications ki backend services ni API dwara access cheyyadaniki use chese service.
Eng: API Gateway is a managed service used to create and manage APIs.

2.What is an API in API Gateway?
An API is a collection of resources, methods, and configurations that define how clients can communicate with backend services.
It provides an endpoint through which applications can send requests and receive responses.
Simple ga gurthupettuko:
API ante client and backend madhya communication kosam create chese interface.
Eng: An API provides an interface for clients to communicate with backend services.

3.What is a Resource in API Gateway?
A Resource represents a URL path in an API.
For example, /users or /orders can be created as resources to represent different parts of an application.
Simple ga gurthupettuko:
Resource ante API lo URL path like /users or /orders.
Eng: A Resource represents a URL path in an API.

4.What is a Method in API Gateway?
A Method defines the HTTP operation that can be performed on a resource.
Common methods are GET, POST, PUT, PATCH, and DELETE.
Simple ga gurthupettuko:
Method ante resource meeda perform chese HTTP operation like GET or POST.
Eng: A Method defines the HTTP operation performed on an API resource.

5.What is a GET Method?
A GET method is commonly used to retrieve data from a backend service.
For example, GET /users can be used to retrieve user information.
Simple ga gurthupettuko:
GET ante backend nundi data retrieve cheyyadaniki use chestham.
Eng: GET is commonly used to retrieve data from a backend service.

6.What is a POST Method?
A POST method is commonly used to send data to a backend service and create a new resource.
For example, POST /users can be used to create a new user.
Simple ga gurthupettuko:
POST ante backend ki data send chesi new resource create cheyyadaniki use chestham.
Eng: POST is commonly used to send data and create a new resource.

7.What is a Stage in API Gateway?
A Stage is a logical environment used to deploy and manage a specific version of an API.
Common examples are dev, test, and prod.
Simple ga gurthupettuko:
Stage ante API ki dev, test, prod laga different environments maintain cheyyadaniki use chestham.
Eng: A Stage is an environment used to deploy and manage an API.

8.Why do we use Stages in API Gateway?
We use stages to maintain different environments of the same API, such as development, testing, and production.
Each stage can have its own configuration and deployment settings.
Simple ga gurthupettuko:
Different environments ni separate ga manage cheyyadaniki Stages use chestham.
Eng: Stages help manage different environments of an API separately.

9.What is an API Gateway Authorizer?
An Authorizer is used to control access to an API by validating the identity or credentials of the client.
It can use services or mechanisms such as IAM, Lambda authorizers, or Amazon Cognito depending on the API type.
Simple ga gurthupettuko:
Authorizer ante API ni access chese user or client valid aa kaada ani check chesi access control cheyyadam.
Eng: An Authorizer validates clients and controls access to an API.

10.What is a Lambda Authorizer?
A Lambda Authorizer is a Lambda function used to validate incoming API requests and determine whether the client should be allowed to access the API.
Simple ga gurthupettuko:
Lambda Authorizer ante request valid aa kaada ani Lambda function dwara check chesi API access ivvadam.
Eng: A Lambda Authorizer uses a Lambda function to validate API requests.

11.What is API Gateway Throttling?
Throttling controls the number of API requests that can be processed within a specific period.
It helps protect backend services from receiving too many requests at the same time.
Simple ga gurthupettuko:
Throttling ante API ki ekkuva requests rakunda request rate ni control cheyyadam.
Eng: Throttling controls the rate of requests sent to an API.

12.Why do we use Throttling?
We use throttling to prevent excessive API traffic and protect backend services from overload.
It can also help maintain stable application performance.
Simple ga gurthupettuko:
Backend overload avvakunda API requests ni limit cheyyadaniki Throttling use chestham.
Eng: Throttling protects backend services from excessive API traffic.

13.What are API Keys?
API Keys are values that can be used to identify API clients and control access to APIs when API key authentication is configured.
They can also be used with usage plans to control API usage.
Simple ga gurthupettuko:
API Key ante API ni use chese client ni identify and usage control cheyyadaniki use chese key.
Eng: API Keys identify API clients and can help control API usage.

14.What is a Usage Plan?
A Usage Plan defines how API clients can access an API and can include API keys, throttling limits, and quotas.
It helps control and manage API usage.
Simple ga gurthupettuko:
Usage Plan ante API client usage ni throttle and quota dwara control cheyyadaniki use chestham.
Eng: A Usage Plan helps control API usage with throttling and quotas.

15.What is an API Gateway Endpoint?
An API Gateway endpoint is the URL through which clients send requests to the API.
The endpoint allows clients to communicate with the configured API resources and methods.
Simple ga gurthupettuko:
API Endpoint ante client request send chese API URL.
Eng: An API Gateway endpoint is the URL used by clients to access an API.

16.What is the difference between Resource and Method?
A Resource represents the URL path, while a Method defines the HTTP operation performed on that resource.
For example, /users is a resource and GET /users is a method.
Simple ga gurthupettuko:
Resource ante URL path, Method ante aa path meeda perform chese operation.
Eng: A Resource is a URL path, while a Method defines the HTTP operation.

17.What is the difference between API Gateway and Lambda?
API Gateway receives and manages API requests, while Lambda runs the backend code in response to those requests.
They are commonly used together to build serverless APIs.
Simple ga gurthupettuko:
API Gateway requests ni handle chestundi, Lambda backend code ni execute chestundi.
Eng: API Gateway manages API requests, while Lambda runs the backend code.

18.How does API Gateway work with Lambda?
The client sends a request to the API Gateway endpoint.
API Gateway receives the request and invokes the configured Lambda function, which processes the request and returns a response.
Simple ga gurthupettuko:
Client → API Gateway → Lambda → Response.
Eng: API Gateway receives the request and invokes Lambda to process it.

19.How would you secure an API Gateway API?
I would use an appropriate authorization mechanism such as IAM, Lambda authorizer, or Amazon Cognito based on the application requirement.
I would also use throttling, HTTPS, and proper access policies to improve API security.
Simple ga gurthupettuko:
API secure cheyyali ante Authorizer + HTTPS + Throttling + proper access policies use chestham.
Eng: I would secure the API using authorization, HTTPS, throttling, and proper access policies.

20.How would you troubleshoot an API Gateway issue?
I would check the API endpoint, resource, method, stage deployment, authorizer, integration configuration, throttling settings, and logs.
I would also check the backend service such as Lambda or EC2 for errors.
Simple ga gurthupettuko:

API problem vaste Endpoint → Resource → Method → Stage → Authorizer → Integration → Backend check chestham.
Eng: For an API Gateway issue, I would check the endpoint, method, stage, authorization, integration, and backend.

21.What is the most important thing to remember about API Gateway?
API Gateway is a managed service used to create, publish, secure, and manage APIs.
The important concepts are APIs, Resources, Methods, Stages, Authorizers, Throttling, and API Keys.
Simple ga gurthupettuko:
API Gateway ante APIs ni create, secure, manage and expose cheyyadaniki use chese AWS service.
Eng: API Gateway is used to create, secure, manage, and expose APIs.